

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	29 June 2026
Name	HARRI VENDHAN L
Project Name	Metro ticket generating system in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

METRO TICKET BOOKING LIFECYCLE

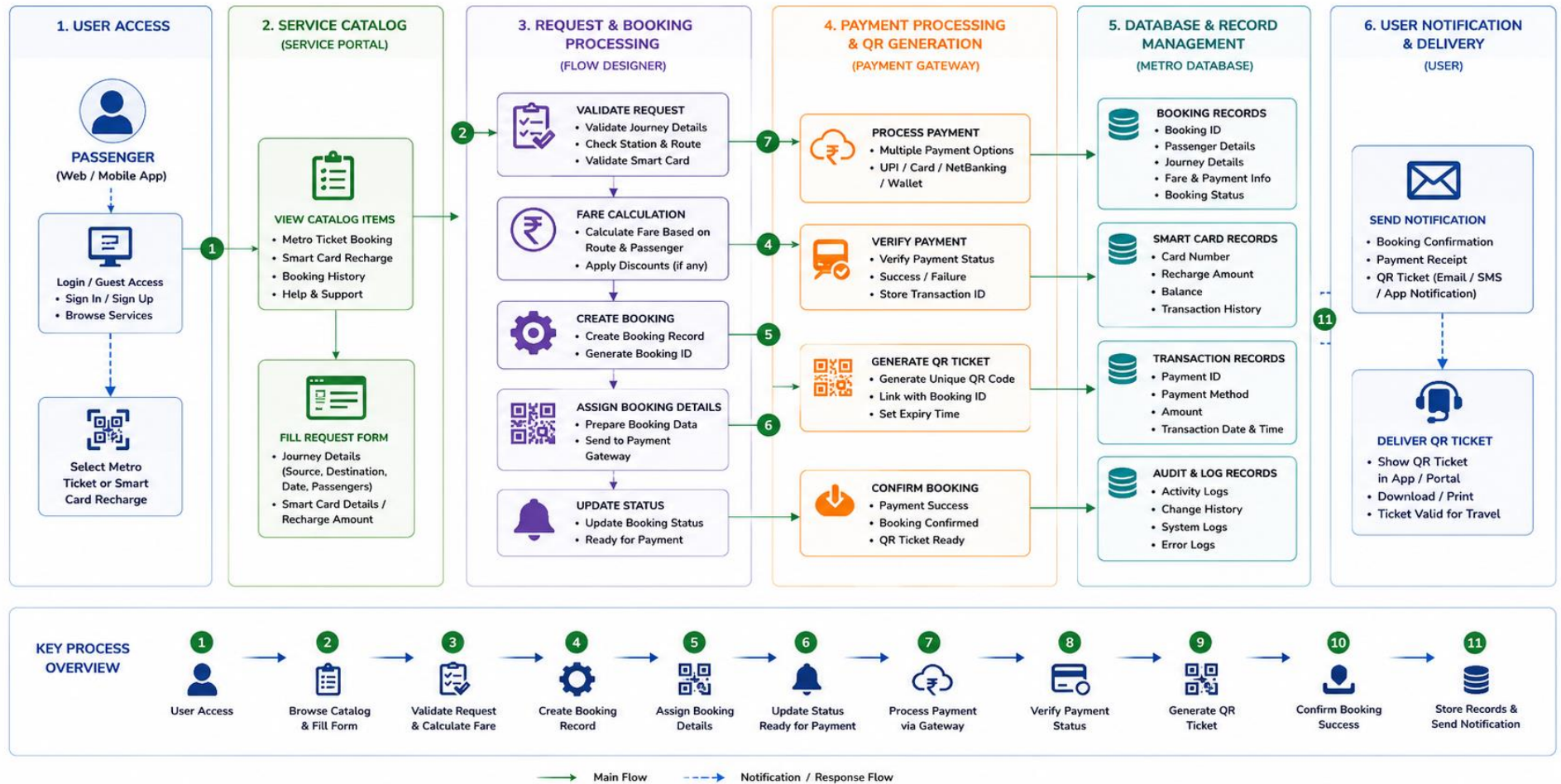


Table-1 : Components & Technologies

S.No.	Component	Description	Technology / Tool
1.	 User Interface (UI)	Provides an intuitive interface for passengers to book tickets, recharge smart cards, and view booking history.	ServiceNow Next Experience UI / Service Portal (HTML5, CSS3, AngularJS)
2.	 Service Catalog	Displays and manages catalog items for Metro Ticket Booking and Smart Card Recharge.	ServiceNow Service Catalog (Catalog Items, Variables, UI Policies)
3.	 Workflow Automation	Automates business processes such as booking creation, fare calculation, payment, and record insertion.	ServiceNow Flow Designer (Record Triggers, Subflows, Actions)
4.	 Business Rules & Validations	Validates user inputs, enforces business rules, and ensures data consistency.	ServiceNow Business Rules, Script Includes, Data Policies
5.	 Database (Operational)	Stores operational data such as bookings, payments, smart cards, stations, routes, and transactions.	ServiceNow Internal Database (MySQL / Oracle)
6.	 Cloud Database	Centralized cloud storage for scalable and secure management of booking, payment and recharge data.	ServiceNow Cloud DB Platform (Requested Items, Catalog Tasks, Payment Records)
7.	 File & Document Storage	Manages attachment files like payment receipts, ID proofs, tickets, and refund documents.	ServiceNow Attachment Engine (Sys_attachment Table & Related Tables)
8.	 Payment Gateway Integration	Handles secure payment transactions and status updates via external payment gateway.	Payment Gateway APIs (UPI, Cards, Net Banking, Wallets)
9.	 External System Integration	Integrates with external metro system services for route data, station info, fare APIs, and QR code services.	REST / SOAP APIs (Route API, Station API, Fare API, QR Code API)
10.	 Analytics & ML Services	Provides analytics, demand prediction, and recommendations for future enhancements.	ServiceNow Predictive Intelligence (Machine Learning Models / AI Search)
11.	 Infrastructure (Cloud / Server)	Hosts the ServiceNow instance and manages scalability, uptime, and performance.	ServiceNow Cloud Infrastructure (SaaS / PaaS), MID Server, Load Balancer, CDN

Table-2 : Application Characteristics

S.No.	Characteristics	Description	Technology / Approach
1.	 Open & Extensible Platform	Built on the ServiceNow platform with open standards and extensible modules for future growth.	ServiceNow Platform, JavaScript, REST APIs, IntegrationHub (Spokes)
2.	 Security & Compliance	Ensures secure access, data encryption, and compliance with security best practices.	RBAC, ACL, Data Encryption (TLS 1.2+), MFA, Security Policies & Audit Logs
3.	 Scalability & Performance	Designed to handle high transaction volumes with optimized workflows and cloud scalability.	ServiceNow Scalable Architecture, Load Balancing, Caching, Asynchronous Processing
4.	 High Availability	Ensures continuous availability and disaster recovery with minimal downtime.	ServiceNow High Availability (HA) Architecture, Active-Active Data Centers, Backup & Failover
5.	 Usability & Accessibility	Provides responsive, user-friendly, and accessible interface across devices.	Next Experience UI, Responsive Design, WCAG 2.1 Standards, Multi-language Support

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>